

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Retail Sale Performance Analysis For Business Decision Making

Project Domain

Microsoft Excel and Power BI

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:



Student Name

Rakshita Srivastava



Student Email

Rakshitasrivastava450@gmail.com



Guide

Ms. Umamaheshawa

Institution:

Uttaranchal University , Dehradun

Year - 2026

ACKNOWLEDGEMENT

We would like to take this opportunity to express our deep sense of gratitude to all individuals who helped us directly or indirectly during this project work.

Firstly, we would like to thank our supervisor Ms. Umamaheshwari & Ms. Surbhi for being a great mentor and the best adviser we could have asked for. Their advice, encouragement, and constructive feedback served as a source of innovative ideas and inspiration, and were central to the successful completion of this project. The confidence shown in us was the biggest source of motivation throughout this journey.

I would also like to extend my sincere and greatest thanks to our mentor Mr. Sagar Sir. Throughout the journey he taught us many new things. He guided us and because of his guidance, support and teachings I was able to overcome my AI fear.

We extend our sincere thanks to the Edunet Foundation, Microsoft, LinkedIn, and SAP for providing this fellowship opportunity, the learning resources, and the platform to develop and showcase our AI skills.

We are also grateful to our institution Uttarakhand University, for providing the necessary infrastructure and support throughout the project duration.

Finally, we thank our family and friends for their continuous encouragement and support.

ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

In this project, an online retail sales dataset was analyzed using Microsoft Excel to identify meaningful business insights. The dataset was first examined and cleaned by checking for missing values, duplicate records, and formatting inconsistencies to ensure data accuracy. After data cleaning in Excel, the dataset was imported into Power BI. Various visualizations such as bar charts, pie charts, line charts, and scatter plots were created. Interactive slicers were added to filter data based on country and date. A dashboard and a detailed table page were developed to present insights effectively. The analysis revealed that the United Kingdom generated the highest revenue, while returns accounted for only a small portion of total transactions. These visualizations helped transform raw data into clear insights that can support better business understanding and decision-making.

TABLE OF CONTENTS

Abstract of the Project

List of Figures

List of Tables

Chapter 1 — Introduction

- 1.1 Problem Statement
- 1.2 Motivation
- 1.3 Objectives
- 1.4 Scope of the Project

Chapter 2 — Literature Survey

Chapter 3 — Proposed Methodology

- 3.1 System Design
- 3.2 Modules / Components Used
- 3.3 Data Flow Diagram (DFD)
- 3.4 Advantages
- 3.5 Requirement Specification

Chapter 4 — Implementation and Results

- 4.1 Data Cleaning Results
- 4.2 Visualization / Output Results
- 4.3 AI Tool Integration Results

Chapter 5 — Discussion and Conclusion

References

Appendices

LIST OF FIGURES

Figure No.	Figure Title / Description	Page No.
Figure 1	Raw Data screenshot	11
Figure 2	Cleaned Dataset screenshot	11
Figure 3	Data Analysis	12
Figure 4	Visualization using charts	12
Figure 5	AI tool (ChatGpt) used for data analysis and insight generation	13
Figure 6	DXD Level 0- context diagrams of retail data analysis system	13

Figure 7	DFD Level 1 – Detailed Process Flow of Retail data analysis	14
Figure 8	Raw data set (before)	16
Figure 9	Cleaned dataset (after)	17
Figure 10	Visualizations	18
Figure 11	ChatGPT Used for Dataset Analysis and Insight Generation	18
Figure 12	Power BI Dashboard	19
Figure 13	Sample of the Raw Online Retail Dataset	23

LIST OF TABLES

Table No.	Table Title / Description	Page No.
Table 1	Review of existing Research Studies	8-9
Table 2	Hardware Requirements	14
Table 3	Software Requirements	15
Table 4	Cleaned Dataset Summary — Issues Found and Fixed	16
Table 5	Summary of Key Findings from Retail Data Analysis	20

CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

Retail industries have a high volume of transactional data being generated every day. This includes data regarding the sales of products, customer purchases, pricing, and regional performance. However, the unprocessed data does not offer much insight for the purpose of decision-making. It is difficult for the organization to derive useful information from the data because it is unstructured and does not have proper visualization techniques applied to it.

One of the biggest challenges faced by retail businesses is the lack of understanding regarding the sales performance in different regions, the products that have the highest demand, and the sales trends. Without proper insights, the business may face problems like ineffective management of products, lack of growth in sales revenue, and inability to differentiate between successful and unsuccessful sales transactions. The lack of proper analysis makes it difficult for the business to differentiate between sales and returns. This project seeks to address these issues by employing data analysis of retail sales data using Microsoft Excel. The project focuses on analyzing the sales data by identifying various patterns, including regional sales performance, product demand, comparing sales and returns, as well as trends over time. This project seeks to demonstrate the importance of data analysis in enhancing business decisions as well as sales performance.

1.2 Motivation

- The goal of this project is to understand how data analytics techniques can be used on real-world retail datasets to produce useful insights. As data-driven decision-making grows in importance, businesses need effective tools to analyze large amounts of sales data and find patterns that can help with strategic planning.
- This project is significant because it offers hands-on experience in managing business transaction data and using analytical techniques with Microsoft Excel. Knowing how to organize, clean, and analyze datasets is an essential skill in today's data analytics and business intelligence fields.

- In practice, similar analysis can help businesses keep track of product demand, spot high-performing regions, and monitor sales trends over time. These insights can lead to better decision-making in areas like marketing strategies, inventory planning, and sales forecasting.
- This project also reflects current industry trends where organizations increasingly depend on data analytics tools to transform raw business data into valuable insights that improve efficiency and overall performance.

1.3 Expected Outcomes

- Identification of top-performing products based on sales volume
- Analysis of regional sales contribution
- Understanding of sales vs returns behavior
- Identification of sales trends over time
- Generation of clear visual insights using charts and tables
- Support for business decision-making through data-driven insights

1.4 Objectives

- To analyze retail sales data for identifying business insights
- To evaluate product performance and demand patterns
- To analyze regional sales contribution
- To visualize sales trends and transaction patterns
- To support business decision-making using data analytics tools
- To analyze retail sales data and create interactive dashboards using Power BI for better business decision-making.

1.5 Scope of the Project

- This project analyses an online retail transaction dataset containing information about products, units sold, sales regions, and transaction values.
- The analysis focuses on understanding sales performance, transaction patterns, and product demand using Microsoft Excel and Power BI.
- The project demonstrates how raw transactional data can be cleaned, organized, and analysed to generate meaningful business insights.
- The tools used in this project include Microsoft Excel for data cleaning and Power BI for data visualization and dashboard creation.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

Paper Title	Author(s)	Year	Technology used	Problem Identified	Methodology
Data-Driven Insights: Enhancing Retail Sales Through Integrating Data Tools for Retail Insights	Janhavi Joshi et al	2025	Data analytics tools, business intelligence techniques	Retail organizations struggle to extract meaningful insights from large sales datasets.	Development of an analytics framework integrating data tools to analyze retail sales performance.
Exploratory Data Analysis of Walmart Outlets Sales using Data Analytics Techniques	Meaga Varsha Ramasami et al.	2023	Data analytics techniques, statistical analysis tools	Understanding sales patterns and product performance in Walmart outlets.	Exploratory Data Analysis (EDA) applied to retail sales datasets to identify trends and patterns.
An Empirical Study on Housing Price Volatility and Retail Sales Growth	Han Jin; Hu Weizhong	2010	Statistical data analysis	Examining the relationship between housing price fluctuations and retail sales growth.	Statistical modeling and Granger causality testing on historical economic data.
Data Presentation and Application of Machine Learning	Natalya V. Razmochaeva; Dmitry M. Klionski	2019	Machine learning algorithms, data	Improving automation and decision-making in retail sales	Machine learning-based analytical models

Methods for Automating Retail Sales Management Processes			visualization tools	management systems.	combined with data presentation techniques.
Comprehensive Analysis of Retail Sales Forecasting Using Machine Learning and Deep Learning Methods	Suresh B. S.; M. Suresh	2023	Machine learning and deep learning models	Accurate prediction of future retail sales and demand forecasting.	Application of predictive models using ML and deep learning algorithms on retail datasets.

Table 1: Review of existing Research Studies

2.2 Tools and Technologies Review

- **Microsoft Excel** - Microsoft Excel was used for data cleaning and preprocessing. The dataset was examined for missing values, duplicate records, and formatting issues. Additional columns such as total transaction value, transaction type, and date fields were created to make the dataset suitable for analysis.
- **Data Analytics Techniques** - Basic data analytics techniques such as data cleaning, sorting, filtering, and exploratory data analysis were applied to understand the structure of the retail dataset and extract meaningful insights.
- **Power BI** - Power BI was used for data visualization and dashboard creation. The cleaned dataset was imported into Power BI, where various interactive charts and visuals were developed. Features such as slicers were used to filter data dynamically based on country and date, enabling better analysis and user interaction.
- **AI Chatbot (ChatGPT)** - AI chatbot tools assisted in generating summaries, explanations, and insights during the analysis and report preparation process. AI prompts helped streamline the documentation and reporting process.

2.3 Gaps Identified

- Many existing studies focus on advanced machine learning and deep learning techniques for retail sales forecasting and automation. However, these approaches often require specialized technical knowledge and complex tools that may not be accessible to beginners or small businesses.
- Some research studies, such as exploratory data analysis of retail sales datasets, focus mainly on identifying patterns but do not emphasize simple visualization techniques that help users clearly understand business insights.
- Certain studies analyze macroeconomic relationships affecting retail sales, such as housing price fluctuations, but they do not focus directly on practical retail transaction analysis for business decision-making.
- Most existing research emphasizes predictive modeling and automated systems rather than simple, practical approaches using widely available tools like Microsoft Excel for analyzing retail sales datasets.
- Therefore, this project focuses on applying basic data analytics techniques using Microsoft Excel to analyze retail transaction data and generate clear business insights through pivot tables and visualizations.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

- **Data Source / Input:** The raw dataset used in this project is the Online Retail transaction dataset stored in CSV format. It contains approximately **701 records and 8 attributes**, including Invoice No, Stock Code, Description, Quantity, Invoice Date, Unit Price, Customer ID, and Country. The dataset represents transactional sales data from a retail store, mainly based in the United Kingdom. This raw dataset was imported into Microsoft Excel for further data cleaning, analysis, and visualization.

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	22752	SET 7 BABUSHKA NESTING BOXES	2	01-12-2010 08:26	7.65	17850	United Kingdom

Figure 1 : Raw data

- **Data Cleaning:** The raw retail dataset was processed in Microsoft Excel to get it ready for analysis. During the cleaning process, we removed duplicate records and filtered out rows with missing values to improve data quality. The data was then organized into a structured format, and some fields were reformatted for consistency. We created additional columns such as Units Sold, Date of Transaction, Time of Transaction, Sales Region, Transaction Type, and Total Transaction Value to simplify analysis. After these steps, we obtained a refined dataset that was used for further analysis through pivot tables and visualizations.

InvoiceNo	Product Code	Description	Units Sold	Date of Transaction	Time of Transaction	UnitPrice	CustomerID	Sales Region	Transaction Type	Total transaction value
536365	85123A	White Hanging Heart T-Light Holder	6	01-12-2010	08:26	₹ 2.55	17850	United Kingdom	Sale	₹ 15.30
536365	71053	White Metal Lantern	6	01-12-2010	08:26	₹ 3.39	17850	United Kingdom	Sale	₹ 20.34
536365	84406B	Cream Cupid Hearts Coat Hanger	8	01-12-2010	08:26	₹ 2.75	17850	United Kingdom	Sale	₹ 22.00
536365	84029G	Knitted Union Flag Hot Water Bottle	6	01-12-2010	08:26	₹ 3.39	17850	United Kingdom	Sale	₹ 20.34
536365	84029E	Red Woolly Hottie White Heart.	6	01-12-2010	08:26	₹ 3.39	17850	United Kingdom	Sale	₹ 20.34
536365	22752	Set 7 Babushka Nesting Boxes	2	01-12-2010	08:26	₹ 7.65	17850	United Kingdom	Sale	₹ 15.30

Figure 2 : Cleaned data

- **Analysis / Processing:** The cleaned dataset was imported into Power BI, where it was analyzed using built-in features to identify key patterns such as regional sales performance, product demand, and sales versus returns. Data was structured and transformed to enable accurate analysis and better visualization of business insights.

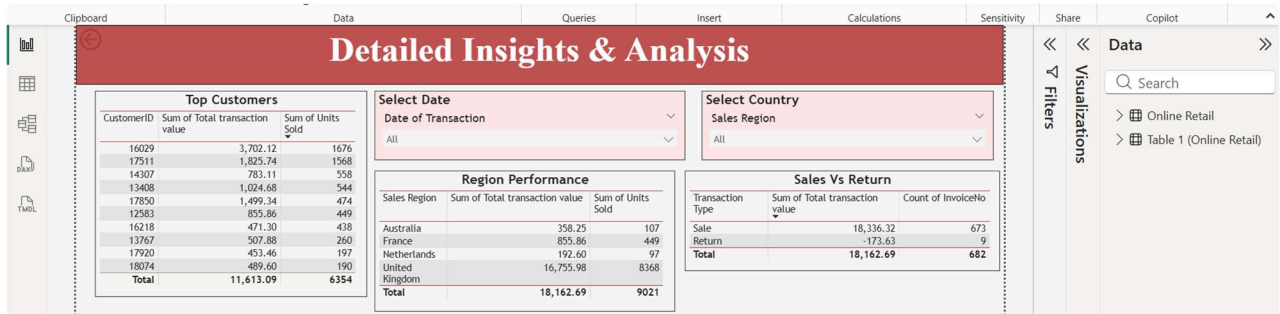


Figure 3 : Data Analysis

- **Visualization / Output:** The results were visualized using an interactive Power BI dashboard. It includes charts such as sales by region, top products, sales trends, and price vs quantity, along with a detailed table page showing region performance, product performance, top customers, and date-wise sales. Slicers were added to filter data by country and date, making the dashboard dynamic and easy to use.

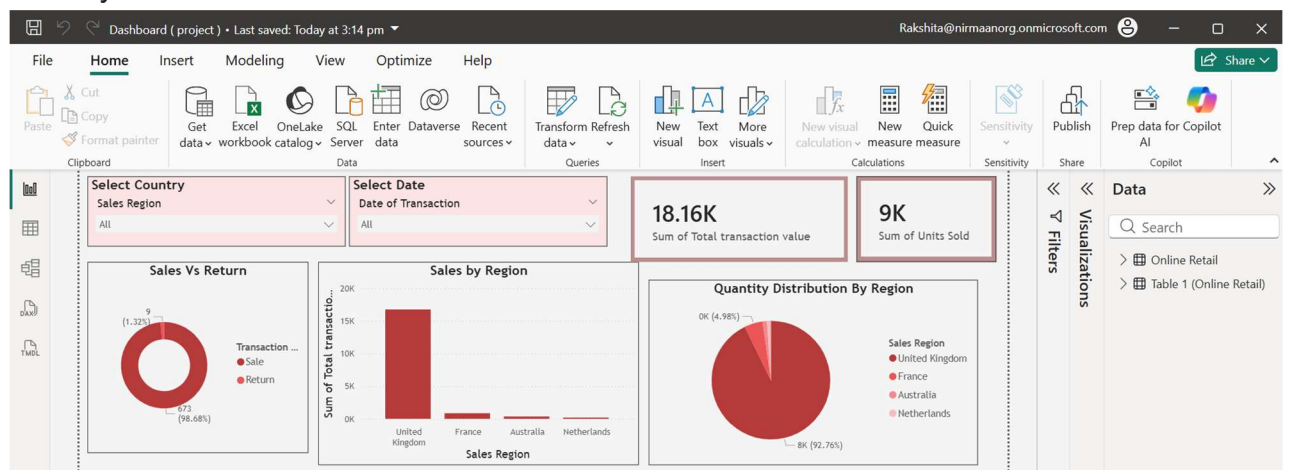


Figure 4 : Visualization using charts

3.2 Modules / Components Used

List the key modules or functional components of your project:

- **Module 1 — Data Ingestion:** Loading the raw dataset into Excel / Power BI.
- **Module 2 — Data Cleaning:** Removing inconsistencies, duplicates, and null values.
- **Module 3 — Data Analysis:** Applying calculations, formulas, AI prompts, or DAX measures.

- Module 4 — Visualization / Report Generation: Building dashboards, charts, or AI-generated reports.
- Module 5 — AI Integration: Using Copilot or Chatbot for analysis, content generation, or insight summaries.

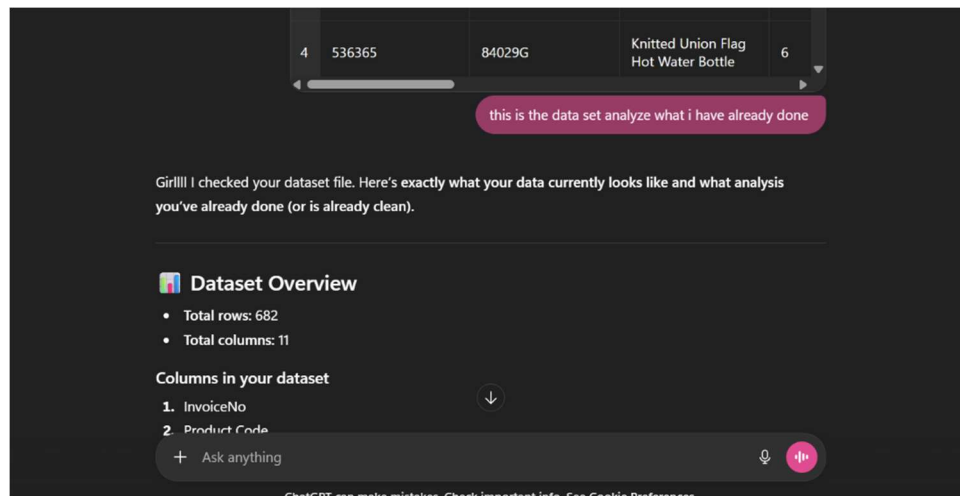


Figure 5 : AI tool (ChatGPT) used for data analysis and insight generation

- Module 6 — Insight Generation: Interpreting the analysed data to identify key patterns such as regional sales performance, top-selling products, and sales trends that support business decision-making.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

Shows the system as a single process with its inputs and outputs.

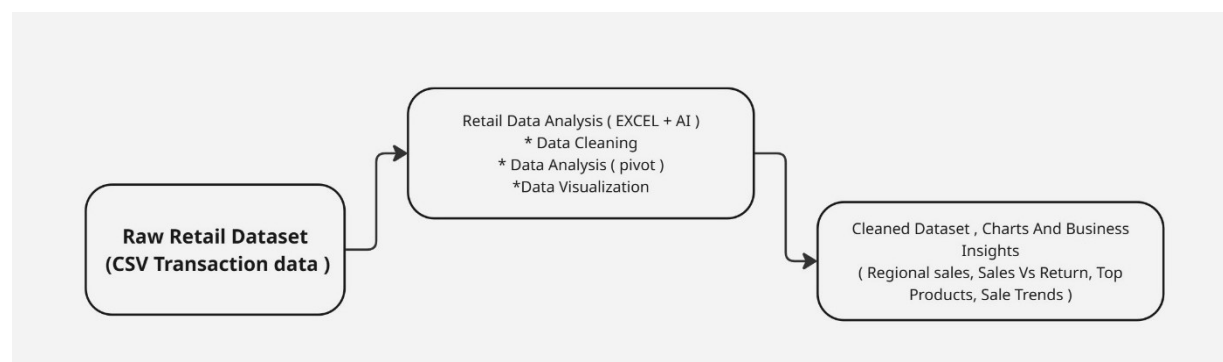


Figure 6 : DFD Level 0- context diagrams of retail data analysis system

DFD Level 1 — Detailed Process Flow

Breaks down the main process into sub-processes. For your project this may include:

- Sub-process 1: Data Loading and Validation
- Sub-process 2: Data Cleaning and Transformation
- Sub-process 3: Analysis and AI Processing
- Sub-process 4: Visualization and Report Generation

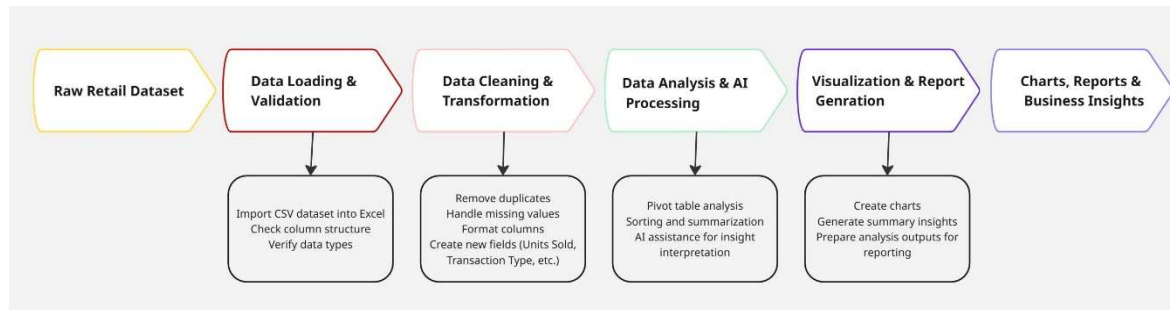


Figure 7 : DFD Level 1 – Detailed Process Flow of Retail data analysis

3.4 Advantages

- **Efficient Data Processing** — The system simplifies the analysis of large retail datasets by organizing and processing transaction data using Microsoft Excel.
- **Improved Data Accuracy** — Data cleaning techniques such as removing duplicates and handling missing values improve the reliability of the dataset.
- **Clear Data Visualization** — Charts and graphs provide clear visual insights into sales patterns, regional performance, and product demand.
- **AI-Assisted Analysis** — AI tools help support data interpretation, generate explanations, and assist in preparing the project documentation.
- **User-Friendly Approach** — The use of widely available tools such as Microsoft Excel makes the system easy to understand and implement without requiring advanced technical expertise.
- **Better Business Insights** — The analysis helps identify important patterns such as top-selling products, sales trends, and regional sales distribution, supporting informed decision-making.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Table 2: Hardware Requirements

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

Table 3: Software Requirements

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Describe the data cleaning process and present the before/after comparison of your dataset.

- **Dataset source:** The dataset used in this project is the Online Retail transaction dataset, which contains records of retail purchases. The dataset was obtained in CSV format and imported into Microsoft Excel for cleaning and analysis.
- **Original dataset size:** The raw dataset contained **701 rows and 8 columns**, including fields such as Invoice No, Stock Code, Description, Quantity, Invoice Date, Unit Price, Customer ID, and Country.
- **Issues found:** During the initial examination of the dataset, several issues were identified including duplicate records, missing values in certain fields (such as Customer ID), and formatting inconsistencies that needed to be corrected before analysis.
- **Cleaning steps applied:** Data cleaning was performed in Microsoft Excel. Duplicate entries were removed, rows containing missing values were filtered out, and column formats were standardized. Additional columns such as Units Sold, Date of Transaction, Time of Transaction, Sales Region, Transaction Type, and Total Transaction Value were created to support easier analysis and visualization.

Issue Identified	Description	Action Taken
Duplicate Records	Some transaction entries appeared more than once	Duplicate rows were removed using Excel tools
Missing Values	Certain records had missing Customer ID or incomplete fields	Rows with missing values were filtered or removed
Formatting Issues	Inconsistent date and numeric formatting	Data types were standardized in Excel
Data Structure	Raw dataset not ready for analysis	Additional columns created for analysis

Table 4 : Cleaned Dataset Summary — Issues Found and Fixed

- **Cleaned dataset size:** After completing the cleaning and transformation steps, the dataset was reduced to **682 rows and 11 columns**, which were used for further analysis and visualization.

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850	United Kingdom

Figure 8 : Raw Dataset (before)

InvoiceNo	Product Code	Description	Units Sold	Date of Transaction	Time of Transaction	UnitPrice	CustomerID	Sales Region	Transaction Type	Total transaction value
536365	85123A	White Hanging Heart T-Light Holder	6	01-12-2010	08:26	₹ 2.55	17850	United Kingdom	Sale	₹ 15.30
536365	71053	White Metal Lantern	6	01-12-2010	08:26	₹ 3.39	17850	United Kingdom	Sale	₹ 20.34
536365	84406B	Cream Cupid Hearts Coat Hanger	8	01-12-2010	08:26	₹ 2.75	17850	United Kingdom	Sale	₹ 22.00
536365	84029G	Knitted Union Flag Hot Water Bottle	6	01-12-2010	08:26	₹ 3.39	17850	United Kingdom	Sale	₹ 20.34
536365	84029E	Red Woolly Hottie White Heart.	6	01-12-2010	08:26	₹ 3.39	17850	United Kingdom	Sale	₹ 20.34
536365	22752	Set 7 Babushka Nesting Boxes	2	01-12-2010	08:26	₹ 7.65	17850	United Kingdom	Sale	₹ 15.30

Figure 9 : Cleaned Dataset (after)

4.2 Visualization / Output Results

- The visualizations were created using Power BI to present key business insights in an interactive format. A bar chart was used to show sales revenue by region, highlighting that the United Kingdom contributes the highest sales compared to other countries.
- A donut chart was used to compare sales and returns, showing that returns form a very small portion of total transactions. A bar chart of top products was created to identify the most demanded products based on units sold.
- A line chart was used to analyze sales trends over time, helping to understand fluctuations in sales performance. Additionally, a scatter chart was used to show the relationship between unit price and quantity sold, indicating that lower-priced products tend to have higher demand.
- Along with these visuals, a detailed table page was created in Power BI to provide deeper insights into region performance, product performance, top customers, and date-wise sales. Interactive slicers were also added to filter the data dynamically based on country and date.

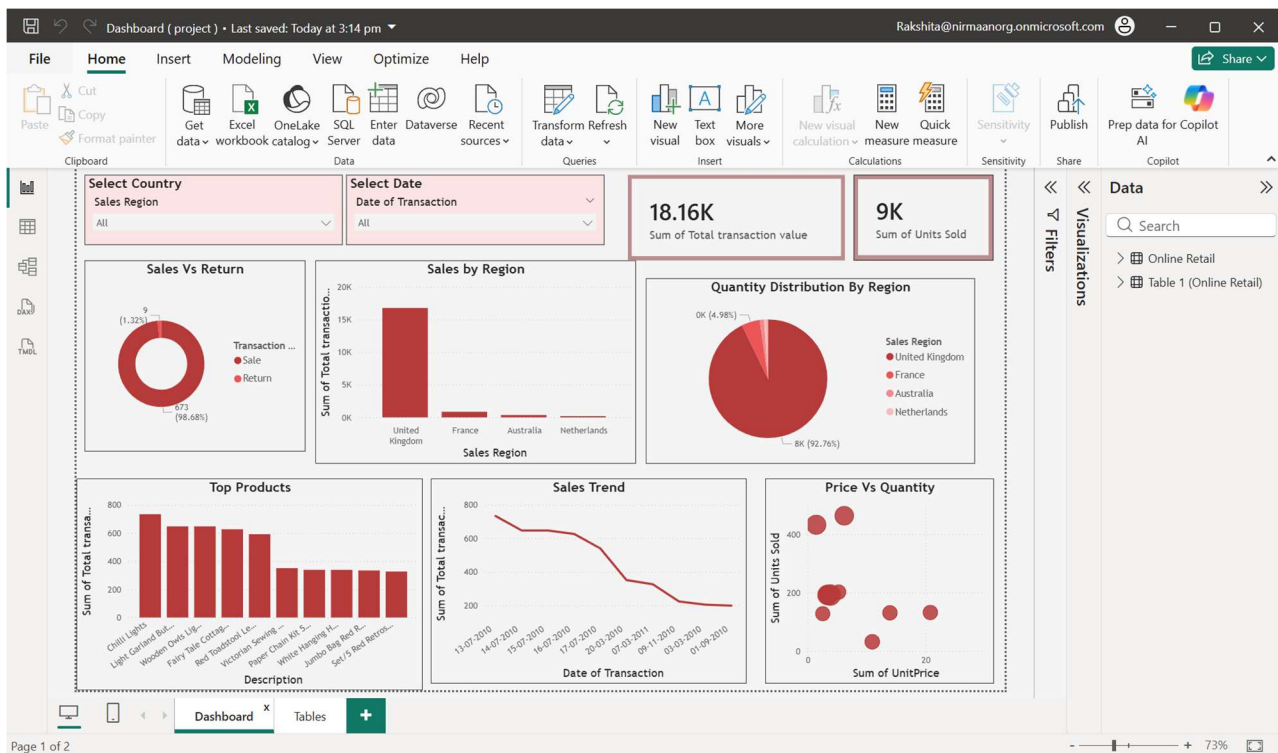


Figure 10 : Visualizations

4.3 AI Tool Integration Results

- **Tool used:** The AI tool used in this project was ChatGPT, which was utilized to assist in analysing the dataset and supporting the preparation of the project documentation.
- **Prompts used:** Some of the key prompts used during the project included:
 1. "Analyse this retail dataset and summarize the structure and columns."
 2. "Suggest insights that can be derived from retail sales data analysis."
 3. "Help generate explanations and documentation for the project report."
- **Output quality:** The AI-generated outputs were helpful in understanding the dataset structure, identifying potential insights, and assisting in writing explanations for the report. The responses were clear, structured, and useful for supporting the analytical process.
- **How it enhanced the project:** Using AI tools helped speed up the analysis and documentation process. It assisted in interpreting the dataset, generating explanations, and organizing insights efficiently, which improved the overall quality and clarity of the project report.

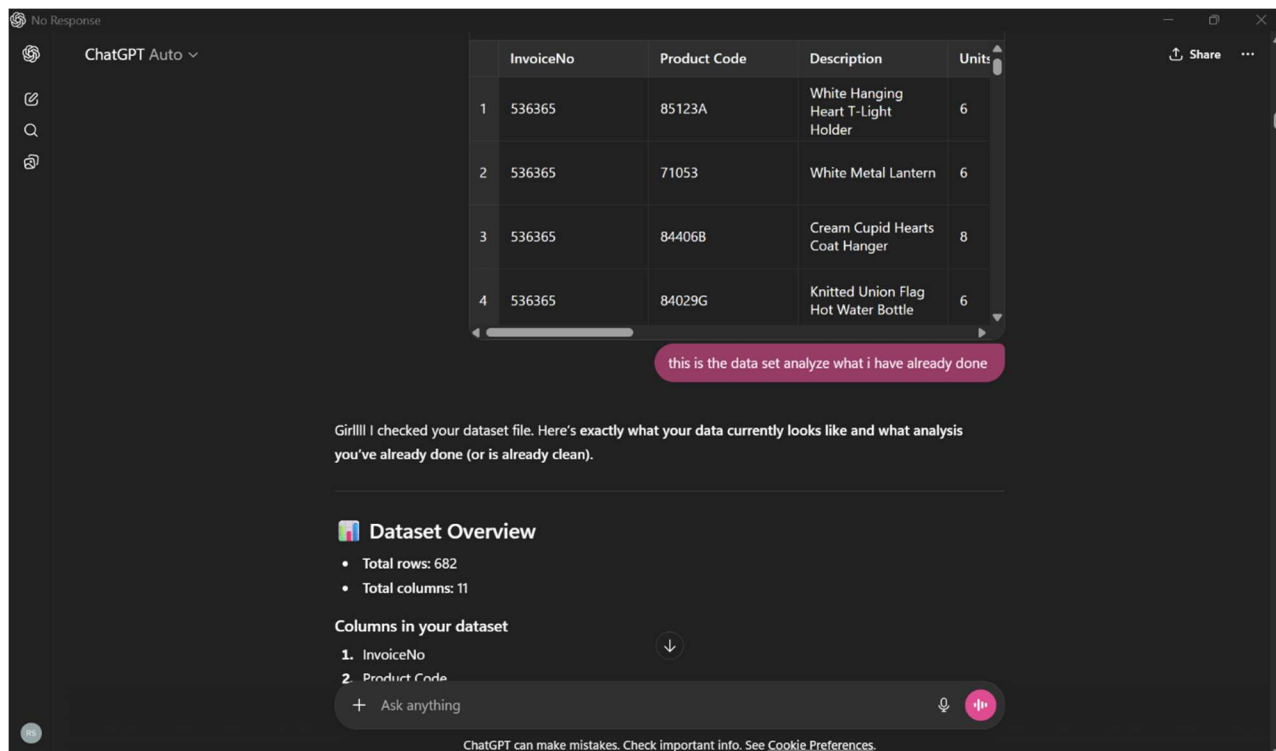


Figure 11: ChatGPT Used for Dataset Analysis and Insight Generation

4.4 Power BI Dashboard

- The Power BI dashboard provides an interactive view of the retail data analysis. It includes key performance indicators such as total sales value and total quantity sold. Visualizations such as sales by region, sales vs return distribution, top products, sales trends over time, and price vs quantity relationship are included. The use of slicers allows users to filter data based on country and date, making the dashboard dynamic and user-friendly for business analysis.

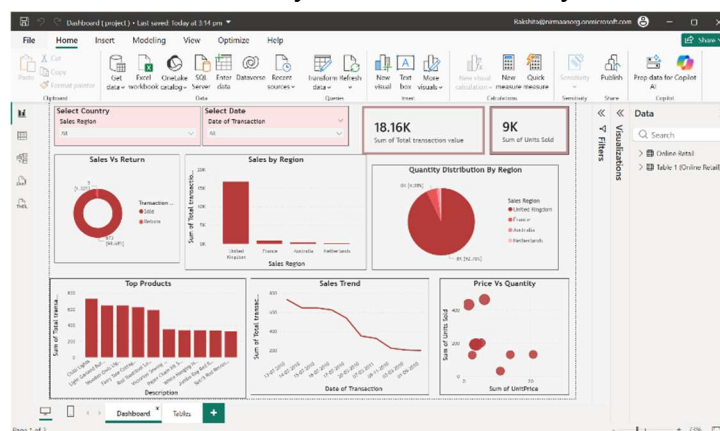


Fig 12 : Power BI dashboard

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- Key Finding 1: After cleaning the raw retail dataset, we removed duplicate records and missing values. This improved the overall quality and reliability of the data used for analysis.
- Key Finding 2: The regional sales analysis showed that the United Kingdom generated the highest revenue compared to countries like France, Australia, and the Netherlands.
- Key Finding 3: The comparison of sales and returns revealed that most transactions were successful sales, while returned items made up only a small portion of the total transaction value.
- Key Finding 4: The analysis of top-selling products identified the items with the highest units sold. This highlighted products that significantly contribute to overall sales performance.
- Key Finding 5: The sales trend visualization helped us identify patterns in transaction values over time. This provided insights into fluctuations in retail sales activity.

Analysis Area	Key Insight
Regional Sales	United Kingdom generated the highest revenue among all countries
Sales vs Returns	Most transactions were successful sales, with returns being minimal
Product Performance	Certain products showed significantly higher demand
Sales Trend	Sales activity showed fluctuations over time

Table 5 : Summary of Key Findings from Retail Data Analysis

5.2 Project Links

Share the relevant links to your project work for evaluation.

- Power BI / Dashboard Link : https://app.powerbi.com/links/x7OmFpxxxP?ctid=aed43286-0f66-47d6-942b-d31a4b8addca&pbi_source=linkShare
- Github Repository Link : <https://github.com/rakshitasrivastava-dotcom/Retail-Sale-Performance-Analysis-For-Business-Decision-Making-.git>

5.3 Limitations

- Limitation 1: The analysis was conducted on a filtered subset of the original retail dataset. This may not fully represent all transaction patterns in the complete dataset.
- Limitation 2: The project primarily focuses on descriptive analysis using Microsoft Excel and Power BI, and does not include advanced predictive modeling techniques.
- Limitation 3: The analysis mainly focuses on descriptive insights like sales distribution and product performance. It does not include more complex predictive modelling techniques.
- Limitation 4: The insights generated depend on the available dataset. They may vary if we consider additional data sources or longer time periods.

5.4 Future Work

- The project can be extended by integrating real-time retail transaction data to enable continuous monitoring of sales performance.
- Advanced analytics techniques such as machine learning models could be applied to predict future sales trends and customer purchasing patterns.
- The analysis could be expanded by incorporating additional datasets from different regions or time periods to generate more comprehensive insights.
- A more interactive dashboard could be developed using tools such as Power BI to allow users to explore the data dynamically.
- AI tools could be further integrated to automate data preprocessing, generate automated reports, and assist in deeper business analysis.

5.5 Conclusion

The project involved the analysis of the online retail data set to obtain useful business insights using Microsoft Excel and Power BI . This included the cleaning of the data by removing duplicate values and handling missing values. After the cleaning of the data, various analytical techniques, such as tables and charts, were applied to analyze the sales performance of different regions, the demand for different products, sales trends, and the difference between sales and returns. From the analysis, useful business insights were obtained, including the prominent role of the United Kingdom in the sales revenue and the identification of the selling products. Thus, the project objectives were achieved, highlighting the capabilities of simple data analytics tools to transform raw data into useful business insights. This project also improved the ability to clean, analyze, and visualize the data, as well as the application of AI tools to support the analytical process. Additionally, Power BI was used to create an interactive dashboard and detailed analytical tables, enhancing the visualization and interpretation of business insights

REFERENCES

- Microsoft Corporation. *Microsoft Excel Documentation*. Available at: <https://support.microsoft.com/en-us/excel>
- Microsoft Corporation. *Power BI Documentation*. Available at: <https://learn.microsoft.com/en-us/power-bi/>
- Microsoft Corporation. *Microsoft Copilot — AI-Powered Productivity*. Available at: <https://copilot.microsoft.com>
- Edunet Foundation. *AICW Fellowship Program Guidelines*. 2026.
- IEEE Xplore Digital Library. *Research Papers on Retail Sales Data Analysis (taken for this project)*. Available at: <https://ieeexplore.ieee.org>
- Joshi, J., Meshram, A., Anarase, D., Dhanorkar, N., & Narkhede, M. *Data-Driven Insights: Enhancing Retail Sales Through Integrating Data Tools for Retail Insights*. IEEE Conference on Computing and Data Science, 2025.
- Ramasami, M. V., Thangaraj, R., Kumar, S. M., & Eswaran, S. *Exploratory Data Analysis of Walmart Outlets Sales Using Data Analytics Techniques*. IEEE International Conference on Digital Applications, 2023.
- Jin, H., & Weizhong, H. *An Empirical Study on Housing Price Volatility and Retail Sales Growth*. IEEE International Conference on Information Management and Engineering, 2010.
- Razmochaeva, N. V., & Klionskiy, D. M. *Data Presentation and Application of Machine Learning Methods for Automating Retail Sales Management Processes*. IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering, 2019.
- Suresh, B. S., & Suresh, M. *A Comprehensive Analysis of Retail Sales Forecasting Using Machine Learning and Deep Learning Methods*. IEEE International Conference on Data Science and Network Security, 2023.
- Online Retail Dataset. Available at: <https://www.kaggle.com/datasets>

APPENDICES

This section provides additional supporting information related to the project that is helpful for understanding the analysis but is not included in the main chapters. It includes a sample of the dataset and important Excel formulas used during the data cleaning and analysis process.

Appendix A — Sample Dataset

- The image below shows a small sample of the original retail dataset used in this project. It contains transaction information such as invoice number, product code, description, quantity, transaction date, unit price, customer ID, and country.

	A	B	C	D	E	F	G	H	I	J
	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country		
2	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850	United Kingdom		
3	536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850	United Kingdom		
4	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850	United Kingdom		
5	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850	United Kingdom		
6	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850	United Kingdom		
7	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	01-12-2010 08:26	7.65	17850	United Kingdom		
8	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	01-12-2010 08:26	4.25	17850	United Kingdom		
9	536366	22633	HAND WARMER UNION JACK	6	01-12-2010 08:28	1.85	17850	United Kingdom		
10	536366	22632	HAND WARMER RED POLKA DOT	6	01-12-2010 08:28	1.85	17850	United Kingdom		
11	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	01-12-2010 08:34	1.69	13047	United Kingdom		
12	536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	01-12-2010 08:34	2.1	13047	United Kingdom		
13	536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	01-12-2010 08:34	2.1	13047	United Kingdom		
14	536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	01-12-2010 08:34	3.75	13047	United Kingdom		
15	536367	22310	IVORY KNITTED MUG COSY	6	01-12-2010 08:34	1.65	13047	United Kingdom		
16	536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	01-12-2010 08:34	4.25	13047	United Kingdom		
17	536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	01-12-2010 08:34	4.95	13047	United Kingdom		
18	536367	22622	BOX OF VINTAGE ALPHABET BLOCKS	2	01-12-2010 08:34	9.95	13047	United Kingdom		
19	536367	21754	HOME BUILDING BLOCK WORD	3	01-12-2010 08:34	5.95	13047	United Kingdom		
20	536367	21755	LOVE BUILDING BLOCK WORD	3	01-12-2010 08:34	5.95	13047	United Kingdom		

Figure 13: Sample of the Raw Online Retail Dataset

Appendix B — Key Excel Formulas / DAX Measures Used

The following Excel formulas and functions were used during the data cleaning and analysis stages of the project:

- =COUNTIF(range, criteria) — used to identify and count duplicate values in the dataset.
- =IFERROR(VLOOKUP(lookup_value, table_array, col_index_num, FALSE), "Not Found") — used to handle lookup errors and ensure consistent output during data validation.
- =SUM(range) — used to calculate total transaction values and summarize numerical data.
- Pivot Table Calculations — Pivot tables were used to summarize the dataset and generate insights such as total sales by region, comparison of sales versus returns, and identification of top-selling products.

Appendix C — AI Prompts Used

The following prompts were used with an AI chatbot (ChatGPT) to assist during the project for data understanding, insight generation, and report preparation.

- Prompt 1: "Analyze this retail dataset and explain the structure and key columns present in the data."
- Prompt 2: "Identify possible insights that can be derived from this retail sales dataset."
- Prompt 3: "Help summarize the results obtained from pivot table analysis of retail sales data."
- Prompt 4: "Assist in writing explanations and documentation for the project report based on the dataset analysis."

Appendix D — Screenshot Gallery

[Insert any additional screenshots, code snippets, or supplementary visuals here that were not included in the main chapters.]